

Question 1

A Generative AI Engineer must create a system that provides multi-turn dialogue capabilities for a customer support chatbot. The chatbot should maintain context across conversations and handle follow-up queries. What tool is essential for maintaining context?

1. A classification model to predict user intents in each query.

Explanation

Incorrect: Classification aids intent identification but does not maintain conversational context.

2. A retriever to fetch relevant past conversations.

Explanation

Incorrect: Retrievers can retrieve content but do not inherently store context dynamically.

3. A memory module integrated into the LLM workflow to store and retrieve past interactions.

Explanation

Correct: Memory modules allow the chatbot to store conversational history, enabling context-aware responses.

4. A summarization model to condense conversation history.

Explanation

Incorrect: Summarization reduces history details, which may compromise context accuracy.

Overall explanation

Memory modules enable dynamic storage and retrieval of past interactions, ensuring continuity and context-aware dialogue.

Question 2

A Generative AI Engineer is designing a pyfunc model to process financial data with pre- and post-processing steps. The model must clean the input, perform predictions, and format results for users. What workflow ensures this functionality?

1. Apply predictions first and handle pre- and post-processing in a separate pipeline.

Explanation

Incorrect: Separating steps reduces efficiency and integration of the workflow.

2. Preprocess financial data (e.g., normalize values), apply the prediction model, and post-process outputs into user-friendly formats like JSON.

Explanation

Correct: This workflow ensures data is clean, predictions are accurate, and results are presented in a structured format for end users.

3. Directly input raw data into the model and rely on post-processing for cleanup.

Explanation

Incorrect: Preprocessing is critical to remove noise and ensure input quality.

4. Skip post-processing and return raw model outputs to users.

Explanation

Incorrect: Raw outputs may lack context and usability for end users.

Overall explanation

Including preprocessing, model inference, and post-processing steps ensures the model processes data effectively and outputs user-ready results.

Question 3

A Generative AI Engineer must evaluate a dataset for legal risks before using it to train a language model. What key factor should they consider?

1. The model architecture being used for training.

Explanation

Incorrect: Model architecture is unrelated to the dataset's legal compliance.

2. The size of the dataset to ensure it meets training requirements.

Explanation

Incorrect: Dataset size is unrelated to legal risks.

3. The presence of sensitive information in the dataset.

Explanation

Incorrect: While important, licensing terms are the primary legal consideration.

4. **Licensing terms that dictate permissible usage and any restrictions on data applications.**

Explanation

Correct: Licensing terms determine how the dataset can be used legally, including restrictions on commercial use or modifications.

Overall explanation

Reviewing licensing terms ensures the dataset is used in accordance with its legal permissions, avoiding potential disputes or violations.

Question 4

A Generative AI Engineer is tasked with creating a prompt for generating job interview questions. The questions should focus on assessing problem-solving skills and teamwork. What is the best prompt structure?

1. **Design a set of job interview questions that evaluate problem-solving and teamwork abilities, providing clear and concise queries.**

Explanation

Correct: This prompt ensures the LLM generates questions tailored to the specific skills being assessed.

2. Summarize common skills assessed during job interviews.

Explanation

Incorrect: Summarization does not generate interview questions.

3. Provide a list of common problem-solving scenarios.

Explanation

Incorrect: Listing scenarios does not generate well-structured questions.

4. Generate general interview questions for job candidates.

Explanation

Incorrect: General prompts do not focus on problem-solving or teamwork skills.

Overall explanation

Explicit prompts that specify the skills and structure needed ensure the generated questions are focused and relevant.

Question 5

A Generative AI Engineer is tasked with designing a RAG system for retrieving technical documentation and generating user-friendly responses. Users report that some responses include irrelevant details. What should the engineer prioritize to improve the system?

1. Use a summarization model to simplify retrieval results.

Explanation

Incorrect: Simplification may still include irrelevant details.

2. Fine-tune the model on technical query data.

Explanation

Incorrect: Fine-tuning improves general performance but does not fix retrieval issues.

3. Increase the model temperature to reduce redundancy.

Explanation

Incorrect: Temperature adjustments do not address retrieval relevance.

4. **Refine the retrieval strategy by filtering results based on relevance scores.**

Explanation

Correct: Filtering retrieval results improves the system's ability to generate focused and relevant responses.

Overall explanation

Refining the retrieval strategy ensures that the system retrieves only the most relevant content, reducing irrelevant details in responses.

Question 6

A Generative AI Engineer has developed an LLM application to answer questions about internal company policies. The engineer must ensure that the application doesn't hallucinate or leak confidential data. Which approach should NOT be used to mitigate hallucination or confidential data leakage?

1. **Fine-tune the model on your data, hoping it will learn what is appropriate and not.**

Explanation

Correct: Fine-tuning cannot reliably mitigate hallucination or prevent data leakage and is not a suitable standalone solution.

2. Limit the data available based on the user's access level.

Explanation

Incorrect: Restricting data access reduces the risk of confidential information leakage.

3. Use a strong system prompt to ensure the model aligns with your needs.

Explanation

Incorrect: System prompts provide explicit instructions to prevent hallucinations and align model behaviour.

4. Add guardrails to filter outputs from the LLM before it is shown to the user.

Explanation

Incorrect: Guardrails effectively filter inappropriate or hallucinated outputs.

Overall explanation

Fine-tuning alone is unreliable for preventing hallucinations or ensuring security. Guardrails, access control, and system prompts are more effective.

Question 7

A Generative AI Engineer must implement guardrails to prevent an AI chatbot from leaking sensitive customer information. What is the most effective approach?

1. Rely on user feedback to detect sensitive data leakage.

Explanation

Incorrect: Relying on feedback is reactive and does not prevent issues during inference.

2. Use prompt-based filters to detect and block sensitive data during inference.

Explanation

Correct: Filters can identify sensitive information and prevent it from being included in responses.

3. Reduce the token length of responses to avoid excessive detail.

Explanation

Incorrect: Reducing response length does not address the risk of sensitive data leakage.

4. Enable temperature-based randomness in outputs.

Explanation

Incorrect: Randomness increases the variability of responses, potentially exposing sensitive data.

Overall explanation

Prompt-based filters actively monitor and block sensitive content, ensuring responses remain secure and compliant with privacy standards.

Question 8

A Generative AI Engineer is integrating a conversational AI model into a web application for restaurant reservations. The system must return concise responses to user queries. How should they optimize the prompt?

1. Use default prompts without modifications.

Explanation

Incorrect: Default prompts may produce longer responses that are not concise.

2. Allow the user to choose response length dynamically.

Explanation

Incorrect: While helpful, dynamic length control does not optimize the default system behavior.

3. Specify in the prompt that the responses should be brief, such as "Provide a short confirmation message for the reservation."

Explanation

Correct: Including specific length-related instructions ensures concise and relevant responses.

4. Rely on the LLM's temperature setting alone to reduce response length.

Explanation

Incorrect: Temperature affects randomness but does not directly control response length.

Overall explanation

Including clear instructions about response length in the prompt ensures the system generates concise and effective messages.

Question 9

A Generative AI Engineer must deploy an endpoint for a RAG application. The endpoint must handle real-time queries and integrate retrieval, embeddings, and LLMs. What sequence of steps should the engineer follow?

1. Skip training embeddings and directly create a vector store.

Explanation

Incorrect: Without embeddings, vector stores cannot represent document semantics.

2. Deploy the endpoint before configuring embeddings.

Explanation

Incorrect: Embeddings must be ready before endpoint deployment.

3. Use retrieval without vector search for simpler implementation.

Explanation

Incorrect: Vector search enhances retrieval quality, making it essential.

4. Train embeddings → Create vector store → Integrate retrieval → Deploy the endpoint.

Explanation

Correct: This sequence ensures all components are ready for real-time query handling.

5. Fine-tune the LLM before deploying the endpoint.

Explanation

Incorrect: Fine-tuning is not necessary for endpoint deployment.

Overall explanation

Proper sequencing ensures an efficient and fully integrated endpoint for real-time RAG applications.

Question 10

A Generative AI Engineer must evaluate the cost-effectiveness of two LLMs for generating responses in a production chatbot. What metrics should they compare? **(Choose two)**

1. Model size in terms of parameters.

Explanation

Incorrect: Model size alone does not directly indicate cost-effectiveness.

2. **Cost per query and latency.**

Explanation

Correct: These metrics provide a direct comparison of operational costs and responsiveness.

3. **Throughput under high-load scenarios.**

Explanation

Correct: Throughput measures the system's capacity to handle queries efficiently at scale.

4. BLEU and perplexity for output quality.

Explanation

Incorrect: These metrics evaluate quality, not cost.

5. Token limit and input size flexibility.

Explanation

Incorrect: These are functional constraints but do not measure cost-effectiveness.

Overall explanation

Comparing cost per query, latency, and throughput ensures a comprehensive evaluation of the cost-effectiveness of LLMs.

Question 11

A Generative AI Engineer must choose an LLM for a customer support chatbot. The chatbot should balance performance and cost while handling common queries effectively. Which metric should guide the selection?

1. Token limit of the LLM.

Explanation

Incorrect: Token limit affects input size but does not balance performance and cost.

2. Model size in terms of parameters.

Explanation

Incorrect: Model size alone does not indicate the balance between cost and performance.

3. **Latency and inference cost per query.**

Explanation

Correct: Latency ensures responsiveness, and inference cost determines economic viability for deployment.

4. BLEU scores for text generation.

Explanation

Incorrect: BLEU evaluates language quality but is not specific to cost or performance trade-offs.

Overall explanation

Latency and inference cost metrics ensure that the LLM meets the performance and cost requirements of a customer support chatbot.

Question 12

LangChain is commonly used for integrating LLMs with retrieval systems. Which key feature of LangChain makes it ideal for this task?

1. Its primary focus on model fine-tuning.

Explanation

Incorrect: LangChain does not handle fine-tuning; it is focused on orchestration.

2. Its use of open-source pre-trained embeddings.

Explanation

Incorrect: While LangChain can leverage embeddings, this is not its defining feature.

3. Its ability to create structured workflows that combine document retrieval and generative capabilities.

Explanation

Correct: LangChain simplifies combining retrieval-based inputs with generative tasks, improving application design.

4. Its built-in visualization tools for output comparison.

Explanation

Incorrect: LangChain is not a visualization tool, and this feature is irrelevant to retrieval integration.

Overall explanation

LangChain's ability to orchestrate workflows between retrieval and generation tasks makes it an excellent choice for Generative AI applications.

Question 13

A Generative AI Engineer must design a system that generates customer response templates based on common queries. The templates should include a greeting, resolution, and follow-up action. What pipeline components are necessary?

1. A classification model to label queries by type.

Explanation

Incorrect: Classification does not generate full response templates.

2. A summarization model to condense user queries.

Explanation

Incorrect: Summarization does not create structured response templates.

3. An LLM to generate templates dynamically based on queries.

Explanation

Correct: LLMs can generate detailed templates tailored to specific queries.

4. A rule-based system to select predefined templates.

Explanation

Incorrect: Rule-based systems lack flexibility for dynamic query handling.

Overall explanation

LLMs dynamically create structured templates with greetings, resolutions, and follow-ups, meeting user requirements.

Question 14

A Generative AI Engineer is tasked with coding a LangChain application to automate querying a product catalog. The system must retrieve product details based on natural language queries. What components are necessary for the chain? **(Choose two)**

1. A summarization model to condense results.

Explanation

Incorrect: Summarization is not needed for querying structured product details.

2. A retriever to fetch product details from the catalog.

Explanation

Correct: Retrievers ensure the chain can access relevant data for natural language queries.

3. A prompt template to structure user queries for the LLM.

Explanation

Correct: Prompt templates ensure consistent and coherent interaction with the model.

4. A pre-trained sentiment analysis model.

Explanation

Incorrect: Sentiment analysis is unrelated to retrieving product details.

5. A rule-based classifier to categorize queries.

Explanation

Incorrect: Rule-based classifiers are too rigid for dynamic product catalog queries.

Overall explanation

Combining retrievers and prompt templates ensures accurate and efficient responses to natural language queries.

Question 15

A Generative AI Engineer is designing a pyfunc model that includes pre- and post-processing for sentiment analysis. What components should the engineer include?

1. Skip the model step and rely only on preprocessing.

Explanation

Incorrect: Preprocessing alone does not perform sentiment analysis.

2. Preprocess input text (e.g., clean text, tokenize), call the model for sentiment prediction, and post-process the output into user-friendly labels.

Explanation

Correct: Pre- and post-processing steps ensure input text is clean and output predictions are formatted for user understanding.

3. Pass raw text to the model without preprocessing.

Explanation

Incorrect: Raw text may include noise, degrading prediction accuracy.

4. Use the model output directly without post-processing.

Explanation

Incorrect: Post-processing converts raw predictions into interpretable results.

Overall explanation

Preprocessing, model inference, and post-processing together ensure high-quality sentiment analysis with interpretable outputs.

Question 16

A Generative AI Engineer must deploy a simple chain using LangChain to enable document retrieval and summarization. Why is LangChain well-suited for this task?

1. It automatically fine-tunes models for retrieval-based tasks.

Explanation

Incorrect: LangChain integrates models but does not handle fine-tuning.

2. It reduces computational costs by optimizing embeddings.

Explanation

Incorrect: LangChain focuses on workflow design, not computational cost reduction.

3. It enhances security for sensitive data by encrypting model outputs.

Explanation

Incorrect: LangChain does not provide built-in security features for model outputs.

4. It provides modular tools for connecting retrievers, embedding models, and language models into a unified pipeline.

Explanation

Correct: LangChain simplifies the creation of end-to-end chains for tasks like retrieval and summarization by offering modular components.

Overall explanation

LangChain's modular tools enable developers to build complete retrieval and summarization pipelines efficiently, making it an excellent choice for such tasks.

Question 17

A Generative AI Engineer must register a pyfunc model that predicts employee attrition to Unity Catalog. The model includes preprocessing for data normalization and postprocessing for interpretable outputs. What steps should the engineer take?

1. Skip logging to MLflow and directly register the model to Unity Catalog.

Explanation

Incorrect: MLflow logging is essential for tracking model details before registration.

2. Skip metadata configuration and use default settings for registration.

Explanation

Incorrect: Metadata provides critical context for effective governance and discoverability.

3. Use MLflow for logging but avoid registering the model to Unity Catalog.

Explanation

Incorrect: Unity Catalog enables centralized governance and compliance tracking.

4. Log the model to MLflow, configure metadata (e.g., schema, version, tags), and register it to Unity Catalog.

Explanation

Correct: Logging and metadata configuration ensure the model is properly tracked and governed within Unity Catalog.

Overall explanation

Logging to MLflow and registering to Unity Catalog ensures the model is properly managed, tracked, and governed.

Question 18

A Generative AI Engineer is tasked with developing an AI tool for generating marketing content. The tool must create product descriptions optimized for e-commerce platforms. What should the prompt include? **(Choose two)**

1. Avoid specifying tone to encourage creativity.

Explanation

Incorrect: Tone guidance ensures consistency and alignment with the platform's branding.

2. Use a generic prompt like "Write a product description."

Explanation

Incorrect: Generic prompts may produce inconsistent results that do not align with e-commerce requirements.

3. Provide few-shot examples of successful descriptions.

Explanation

Correct: Examples help the model generate descriptions aligned with proven e-commerce strategies.

4. Specify product details, tone, and length. Example: "Create a description for a smartwatch. Tone: professional. Length: 100 words."

Explanation

Correct: Including product details, tone, and length ensures the output is optimized for the platform's needs.

Overall explanation

Combining clear instructions (tone, length, details) with examples ensures product descriptions are tailored to the platform and resonate with the target audience.

Question 19

Which indicator should be considered to evaluate the safety of the LLM outputs when qualitatively assessing LLM responses for a translation use case?

1. The accuracy and relevance of the responses

Explanation

Correct: Accuracy and relevance are key indicators of safety, ensuring that the outputs meet the task requirements without errors or harmful content.

2. The similarity to the previous language

Explanation

Incorrect: Safety is not about maintaining linguistic similarity but ensuring relevance and appropriateness of translations.

3. The ability to generate responses in code

Explanation

Incorrect: Generating code is unrelated to evaluating safety in translation tasks.

4. The latency of the response and the length of text generated

Explanation

Incorrect: Latency and text length are performance metrics, not safety indicators.

Overall explanation

Accurate and relevant outputs ensure safe and effective responses in translation use cases by aligning with the user's needs and context.

Question 20

A Generative AI Engineer is tasked with chunking a lengthy novel for a RAG application. The novel includes chapters, dialogues, and descriptive passages. Which chunking strategy will optimize retrieval quality?

1. Split the text by chapters and further divide each chapter into sections based on paragraphs.

Explanation

Correct: Dividing by chapters and paragraphs maintains logical structure and adheres to token limits.

2. Split the text into chunks of a fixed token size (e.g., 500 tokens).

Explanation

Incorrect: Fixed-size chunks risk breaking context, reducing coherence.

3. Include the entire novel as one chunk for context preservation.

Explanation

Incorrect: Entire novel chunks exceed token limits and reduce processing feasibility.

4. Overlap chunks with a 50% overlap to ensure content redundancy.

Explanation

Incorrect: Excessive overlap can create redundant data without improving retrieval quality.

Overall explanation

Logical chunking by chapters and paragraphs ensures meaningful content for retrieval and alignment with model constraints.

Question 21

A Generative AI Engineer is designing a system that converts job descriptions into candidate evaluation criteria. The system must generate a structured checklist for each job description. What components should be included in the pipeline?

1. Use a summarization model to condense job descriptions into shorter forms.

Explanation

Incorrect: Summaries do not translate into evaluation criteria.

2. Use a classification model to label job descriptions by category.

Explanation

Incorrect: Classification does not generate detailed evaluation checklists.

3. Use an embedding model for job descriptions and an LLM to generate evaluation criteria.

Explanation

Correct: Embeddings capture semantic meaning, and LLMs generate structured outputs from descriptions.

4. Use a rule-based system to extract criteria.

Explanation

Incorrect: Rule-based systems lack flexibility for diverse job descriptions.

Overall explanation

Combining embeddings and LLMs ensures the pipeline generates accurate and structured evaluation criteria.

Question 22

A Generative AI Engineer must implement guardrails for a chatbot handling sensitive customer data. What is the most critical step?

1. Filter responses to detect and block any potential private data leakage.

Explanation

Correct: Filtering prevents sensitive information from being included in chatbot responses, ensuring privacy.

2. Enable unrestricted response generation.

Explanation

Incorrect: Unrestricted responses increase the risk of data leakage.

3. Shorten the responses to avoid excessive detail.

Explanation

Incorrect: Short responses do not inherently prevent data leakage.

4. Train the chatbot on smaller datasets.

Explanation

Incorrect: Dataset size does not address privacy concerns directly.

Overall explanation

Filtering responses for private data ensures compliance with privacy standards and prevents sensitive data exposure.

Question 23

An AI model generates recommendations for customer queries but frequently suggests irrelevant products. What adjustment should an engineer prioritize?

1. Add more parameters to the model.

Explanation

Incorrect: Adding parameters does not directly improve query-product alignment.

2. Reduce response length to avoid irrelevant recommendations.

Explanation

Incorrect: Shorter responses do not guarantee improved relevance.

3. Improve embeddings to better capture query-product semantic similarity.

Explanation

Correct: Improving embeddings ensures better alignment between customer queries and product suggestions.

4. Increase the model's temperature during inference.

Explanation

Incorrect: Increasing temperature adds randomness, which may worsen relevance.

Overall explanation

High-quality embeddings improve the semantic alignment between queries and products, addressing issues of irrelevance in recommendations.

Question 24

A Generative AI Engineer must deploy a LangChain-based application for retrieving legal documents and generating summaries. What is the primary role of LangChain in this system?

1. Automating the fine-tuning of LLMs.

Explanation

Incorrect: LangChain does not fine-tune models but integrates pre-trained ones.

2. Managing cloud resources for scalable deployment.

Explanation

Incorrect: LangChain focuses on workflows, not resource management.

3. Visualizing data flows in the application.

Explanation

Incorrect: LangChain is not a visualization tool.

4. Orchestrating the workflow by connecting retrievers, LLMs, and preprocessing steps.

Explanation

Correct: LangChain simplifies the orchestration of complex workflows, such as retrieval and summarization.

Overall explanation

LangChain connects retrievers, preprocessors, and LLMs in a unified pipeline, making it ideal for building modular retrieval and summarization workflows.

Question 25

A Generative AI Engineer is deploying a LLM-based RAG application for financial document analysis. The application must minimize costs while processing lengthy documents. Which strategies are most effective? (Choose three)

1. Optimize token usage by batching multiple queries into a single request.

Explanation

Correct: Batching reduces the number of API calls or inference runs, improving cost efficiency.

2. Reduce the retrieval set size to minimize processing overhead.

Explanation

Incorrect: Reducing retrieval size may compromise the quality of retrieved results.

3. Increase context length to maximize token usage for each query.

Explanation

Incorrect: Increasing context length can lead to higher token usage and increased costs.

4. Use a smaller LLM with high context length support to handle long inputs efficiently.

Explanation

Correct: Smaller LLMs with sufficient context length reduce computational demands, lowering costs while maintaining performance.

5. Enable inference logging to monitor query costs.

Explanation

Correct: Inference logging provides insights into cost drivers, enabling optimization strategies.

Overall explanation

Combining smaller models, batching, and inference logging ensures cost-effective performance for processing lengthy financial documents.

Question 26

A Generative AI Engineer must protect a generative application from adversarial user inputs designed to crash the system. What is the best defensive strategy?

1. Implement input validation rules to check for unacceptable patterns and formats.

Explanation

Correct: Input validation prevents adversarial inputs from compromising system stability and security.

2. Lower the model's response complexity to limit crash risks.

Explanation

Incorrect: Reducing response complexity does not address input-based vulnerabilities.

3. Monitor user inputs and log suspicious activity.

Explanation

Incorrect: Monitoring helps detect issues but does not prevent adversarial inputs.

4. Train the model on a dataset designed to handle adversarial inputs.

Explanation

Incorrect: Training alone cannot prevent all forms of malicious input attacks.

Overall explanation

Input validation rules ensure the application can handle adversarial inputs securely, protecting system stability.

Question 27

A Generative AI Engineer has developed a RAG application designed to answer questions about a series of fantasy novels discussed on the author's web forum. The novels have been chunked and embedded into a vector store containing metadata like page numbers, chapter titles, and book names. User queries retrieve relevant chunks, which are then processed by an LLM for generating responses. Initially, the engineer used intuition to select chunking strategies and configurations but now wants a systematic approach to optimize these settings. Which TWO approaches should the engineer adopt to refine the chunking strategy and parameters?

(Choose two)

1. Create an LLM-as-a-judge metric to assess how well previous questions are answered by specific chunks. Use this metric to optimize chunking parameters.

Explanation

Correct: An LLM-as-a-judge approach allows the engineer to systematically evaluate and refine chunk relevance and sizes.

2. Change embedding models and compare performance.

Explanation

Incorrect: Experimenting with embedding models improves semantic alignment but does not directly optimize chunking strategies.

3. Choose an appropriate evaluation metric (e.g., recall or NDCG) and test chunking strategies such as splitting chunks by paragraphs or chapters. Select the strategy with the highest performance metric.

Explanation

Correct: Using evaluation metrics provides a structured way to identify the most effective chunking strategies.

4. Pass known questions and best answers to an LLM and instruct it to identify optimal token counts. Use a summary statistic (mean, median, etc.) to determine chunk sizes.

Explanation

Incorrect: LLM-driven token suggestions lack a structured approach for robust chunk optimization.

5. Add a classifier for user queries that predicts which book will best contain the answer. Use this to filter retrieval.

Explanation

Incorrect: While helpful for narrowing retrieval, a classifier does not optimize the chunking strategy itself.

Overall explanation

A systematic evaluation using metrics like recall and NDCG, combined with LLM-as-a-judge metrics, ensures the best chunking strategy is selected for optimal performance.

Question 28

A Generative AI Engineer is tasked with fine-tuning a model to identify key topics in technical documents. The dataset contains unnecessary metadata such as timestamps and author names. How should the engineer handle preprocessing?

1. Retain metadata to provide context.

Explanation

Incorrect: Retaining metadata can confuse the model and degrade topic identification performance.

2. Exclude all numerical data, including timestamps.

Explanation

Incorrect: Some numerical data, like dates, may be relevant and should not be excluded without consideration.

3. Process the dataset without filtering.

Explanation

Incorrect: Ignoring irrelevant metadata risks degrading the system's overall performance.

4. Remove all irrelevant metadata during preprocessing.

Explanation

Correct: Removing unnecessary metadata reduces noise, helping the model focus on the actual document content.

Overall explanation

Removing metadata ensures the focus remains on relevant document content, enhancing topic identification accuracy.

Question 29

A Generative AI Engineer must control access to a deployed model serving endpoint. What techniques can they use to ensure secure access?

1. Implement API key authentication and role-based access controls (RBAC).

Explanation

Correct: API keys and RBAC ensure secure and granular access control to model serving endpoints.

2. Restrict access to specific IP addresses but skip authentication.

Explanation

Incorrect: IP restrictions alone do not provide robust authentication mechanisms.

3. Use a public endpoint with a shared access key for all users.

Explanation

Incorrect: Shared keys compromise security and do not provide individualized access controls.

4. Deploy the model on a private network without additional access controls.

Explanation

Incorrect: A private network alone does not ensure endpoint security for external integrations.

Overall explanation

API keys and RBAC provide robust security, ensuring only authorized users can access the deployed model endpoint.

Question 30

A Generative AI Engineer must deploy a PyFunc model that preprocesses input text by cleaning noisy characters and post-processes the output by formatting the result into a JSON structure. Where should the preprocessing and post-processing steps be implemented?

1. Using a data preprocessing library such as Pandas.

Explanation

Incorrect: While Pandas is useful for data transformation, it cannot directly integrate with PyFunc model workflows.

2. Inside the PyFunc wrapper's predict function, as part of the model logic.

Explanation

Correct: PyFunc models are designed to encapsulate pre- and post-processing within the predict function, ensuring all operations are performed seamlessly within the model workflow.

3. At the deployment server level via a custom API wrapper.

Explanation

Incorrect: While feasible, placing these operations at the server level complicates maintenance and debugging.

4. As standalone Python scripts outside the PyFunc model.

Explanation

Incorrect: External scripts add unnecessary complexity and risk inconsistency during deployment.

Overall explanation

PyFunc models allow custom pre- and post-processing logic to be encapsulated within the predict function, ensuring streamlined and integrated workflows during inference.

Question 31

A Generative AI Engineer is designing an LLM-powered tool for summarizing legal contracts. The model often generates hallucinated clauses. What solution should they implement?

1. Apply a summarization model pre-trained on legal datasets.

Explanation

Incorrect: Pre-trained models help improve quality but do not inherently prevent hallucinations.

2. Use a classification model to validate generated summaries.

Explanation

Incorrect: Classification models do not address the source of hallucinations directly.

3. Write a metaprompt instructing the model to generate summaries only based on the provided text.

Explanation

Correct: Metaprompts help constrain the model's outputs, reducing the risk of hallucination.

4. Increase the context length to capture more document content.

Explanation

Incorrect: Increasing context length may not prevent hallucinations and can increase computational costs.

Overall explanation

Metaprompts guide the LLM to focus strictly on source content, reducing the likelihood of hallucinations in contract summaries.

Question 32

A RAG system retrieves irrelevant historical data during queries. How should the engineer improve the relevance of retrieved documents?

1. Retain all documents regardless of age.

Explanation

Incorrect: Retaining all documents, including outdated ones, reduces overall relevance and accuracy.

2. Apply timestamp-based filtering to exclude outdated documents.

Explanation

Correct: Timestamp filtering ensures the system prioritizes recent and contextually relevant content, improving accuracy.

3. Increase the document chunk size to improve coverage.

Explanation

Incorrect: Larger chunks may include irrelevant information, reducing retrieval relevance.

4. Filter based on keyword matches in queries.

Explanation

Incorrect: Keyword-based filtering may miss contextually relevant documents due to synonym usage or rephrased content.

Overall explanation

Timestamp-based filtering ensures retrieved content remains up-to-date and relevant, addressing issues of outdated information.

Question 33

A Generative AI Engineer is tasked with designing a system for generating personalized workout routines. The system should consider user goals, preferences, and activity history. What is the most effective pipeline design?

1. Use a classification model to categorize user preferences.

Explanation

Incorrect: Categorization does not generate personalized workout plans.

2. Use a rule-based system to map inputs to predefined workouts.

Explanation

Incorrect: Rule-based systems lack flexibility for diverse user inputs.

3. Use an LLM to generate workout routines based on inputs, supported by a retriever for user activity history.

Explanation

Correct: Combining an LLM and a retriever ensures dynamic, personalized outputs tailored to user preferences and goals.

4. Use embeddings to represent user inputs semantically.

Explanation

Incorrect: Embeddings alone do not generate actionable plans.

Overall explanation

A combination of an LLM and retriever ensures personalized and dynamic workout routines based on user inputs.

Question 34

A Generative AI Engineer must select an LLM for a question-answering system about medical guidelines. The system needs high accuracy and explainable outputs. Which LLM attribute should they prioritize?

1. Pre-trained embeddings on generic text.

Explanation

Incorrect: Generic embeddings lack the domain-specific focus necessary for medical accuracy.

2. Model size and domain-specific fine-tuning.

Explanation

Correct: Domain-specific fine-tuning ensures high accuracy for medical guidelines, and explainable outputs require a focused model.

3. High perplexity to ensure diverse responses.

Explanation

Incorrect: High perplexity introduces randomness, reducing reliability in medical contexts.

4. Response speed and low latency.

Explanation

Incorrect: Speed is secondary to accuracy and explainability in medical applications.

Overall explanation

Prioritizing domain-specific fine-tuning ensures that the LLM delivers accurate and explainable outputs for medical guideline queries.

Question 35

A Generative AI Engineer is designing a conversational agent using Foundation Model APIs. The system must handle variable query loads efficiently. How can the engineer optimize performance?

1. Use autoscaling to dynamically adjust resources based on query load.

Explanation

Correct: Autoscaling ensures resource availability matches workload demands, maintaining performance.

2. Reduce API rate limits to restrict user queries.

Explanation

Incorrect: Rate limiting does not optimize system resource utilization.

3. Skip monitoring performance metrics to simplify deployment.

Explanation

Incorrect: Monitoring metrics is critical for detecting and addressing performance issues.

4. Fix resource allocation at the maximum expected load.

Explanation

Incorrect: Fixed allocation leads to underutilization during low load and bottlenecks during spikes.

Overall explanation

Autoscaling dynamically adjusts resources to match workload demands, ensuring consistent and efficient performance.

Question 36

A Generative AI Engineer is tasked with preparing a large technical document for a RAG system. The document contains appendices, tables of contents, and redundant footnotes that are irrelevant to the task. What preprocessing step should the engineer take?

1. Merge all sections of the document into a single chunk.

Explanation

Incorrect: Merging content reduces clarity and risks exceeding token limits.

2. Remove irrelevant sections like appendices and tables of contents during preprocessing.

Explanation

Correct: Removing extraneous sections reduces noise and improves retrieval performance by focusing on relevant content.

3. Keep the entire document intact for completeness.

Explanation

Incorrect: Retaining irrelevant sections increases retrieval noise and degrades performance.

4. Highlight irrelevant sections to flag them but keep them in the document.

Explanation

Incorrect: Flagging does not improve system performance; irrelevant content still occupies space.

Overall explanation

Preprocessing ensures the removal of irrelevant sections, enhancing the quality of document retrieval for RAG applications.

Question 37

A marketing agency needs an AI tool to generate engaging headlines for advertisements. The headlines should target specific demographics like "young professionals" or "retirees." What should the prompt include for best results? (Choose two)

1. Provide a few-shot example of successful headlines.

Explanation

Correct: Examples guide the AI by showcasing high-performing outputs.

2. Avoid specifying tone to encourage creativity.

Explanation

Incorrect: Lack of tone guidance risks inconsistency and misalignment with target demographics.

3. Specify the tone, target audience, and desired outcome. Example: "Write a headline targeting young professionals. Tone: professional and motivating. Outcome: highlight career growth opportunities."

Explanation

Correct: Including tone, audience, and goals ensures the AI generates headlines tailored to campaign needs.

4. Use a generic prompt like "Write engaging headlines for ads."

Explanation

Incorrect: Generic prompts may produce irrelevant or inconsistent headlines.

Overall explanation

Combining clear instructions (tone, audience, goals) with examples ensures the AI generates consistent, high-quality headlines that resonate with the intended audience.

Question 38

A Generative AI Engineer must build a QA system for a retail company. The system should retrieve answers about product inventory and availability. What tools should the pipeline include?

1. A retriever to fetch inventory data and an LLM to generate responses.

Explanation

Correct: Retrievers ensure relevant inventory data is fetched, and the LLM dynamically crafts accurate responses.

2. An embedding model to represent inventory data semantically.

Explanation

Incorrect: Embeddings alone do not generate query responses.

3. A summarization model to condense product descriptions.

Explanation

Incorrect: Summarization does not address retrieval or inventory queries.

4. A classification model to categorize products.

Explanation

Incorrect: Categorization does not fulfill the need for dynamic question answering.

Overall explanation

A combination of retrievers and LLMs ensures accurate answers to inventory-related queries, meeting user requirements.

Question 39

A Generative AI Engineer is integrating an LLM into a sales analytics platform. The LLM must suggest actions based on customer purchase patterns. Which feature is critical for success?

1. Fine-tuning on purchase history data for better predictions.

Explanation

Correct: Fine-tuning ensures the LLM aligns with domain-specific data, improving its recommendation accuracy.

2. Choosing the largest available LLM for broader capabilities.

Explanation

Incorrect: Larger models are unnecessary if they are not domain-specific.

3. Relying solely on embeddings without inference capabilities.

Explanation

Incorrect: Embeddings alone do not provide actionable recommendations.

4. Using a default pre-trained LLM without modification.

Explanation

Incorrect: Default models are not specialized for purchase pattern analysis.

Overall explanation

Fine-tuning on domain-specific data ensures the LLM generates actionable insights aligned with customer purchase patterns.

Question 40

A Generative AI Engineer must design a prompt to extract meeting summaries. Each summary should include action items, participants, and decisions made. What is the best prompt design?

1. Highlight the key points of the meeting in a bullet-point format.

Explanation

Incorrect: Bullet points alone may not include all required fields.

2. Provide an overview of the meeting discussion.

Explanation

Incorrect: Overviews lack detail and structure for actionable summaries.

3. Summarize the meeting transcript into a concise report.

Explanation

Incorrect: Concise reports may omit essential elements like action items or participants.

4. Extract a meeting summary including action items, participants, and decisions made, based on the transcript.

Explanation

Correct: This prompt explicitly defines required elements, ensuring the model generates structured, complete summaries.

Overall explanation

Explicit prompts defining required fields ensure the model generates structured, actionable meeting summaries.

Question 41

A Generative AI Engineer is optimizing an LLM-based system for summarizing academic papers. The summaries often miss key points from the results section. What strategy should they use?

1. Summarize without prompts to save computation.

Explanation

Incorrect: Lack of prompts results in unstructured and unreliable summaries.

2. Include section-specific instructions in the prompt, such as "Focus on the results section."

Explanation

Correct: Targeted instructions ensure the model emphasizes important parts of the document.

3. Use vague prompts like "Summarize this paper."

Explanation

Incorrect: Vague prompts fail to direct the model to prioritize key sections.

4. Include the entire paper in the prompt for context.

Explanation

Incorrect: Overloading the prompt with unnecessary context can reduce performance and increase costs.

Overall explanation

Adding section-specific instructions ensures the model focuses on critical information, such as the results section, improving summary relevance.

Question 42

A Generative AI Engineer is evaluating retrieval performance for a RAG application using scientific documents. Which metrics are best suited for this evaluation?

1. Precision and recall to measure relevance and coverage of retrieved results.

Explanation

Correct: Precision ensures retrieved documents are relevant, while recall measures how many relevant documents are retrieved.

2. BLEU to measure textual similarity.

Explanation

Incorrect: BLEU evaluates text similarity in tasks like translation, not retrieval.

3. Token usage to evaluate computational costs.

Explanation

Incorrect: Token usage is unrelated to retrieval quality.

4. Latency to measure speed of retrieval.

Explanation

Incorrect: Latency measures performance but not relevance or coverage.

Overall explanation

Precision and recall are key metrics for evaluating the relevance and completeness of retrieval performance in a RAG application.

Question 43

A Generative AI Engineer needs to generate embeddings for a question-answer retrieval system. User queries are short, while documents are long. Which embedding model strategy is most effective?

1. Use a dual encoder with separate embeddings for queries and documents.

Explanation

Correct: Dual encoders optimize retrieval by aligning short queries with longer document representations.

2. Rely on pre-trained embeddings without customization.

Explanation

Incorrect: Pre-trained embeddings may not align with the specific domain or task requirements.

3. Use a single encoder for both queries and documents.

Explanation

Incorrect: Single encoders are less effective for handling significant differences in input length.

4. Use a classification model instead of an embedding model.

Explanation

Incorrect: Classification models are not designed for retrieval tasks.

Overall explanation

A dual encoder ensures that both short queries and long documents are effectively represented, improving retrieval performance.

Question 44

A Generative AI Engineer is tasked with identifying problematic text in a dataset feeding a RAG application. Some documents contain inappropriate language. What mitigation strategy should they implement?

1. Allow all documents to be ingested and rely on output monitoring.

Explanation

Incorrect: Post-ingestion monitoring does not prevent problematic content from affecting the system.

2. Use a filtering tool to detect and remove inappropriate language before ingestion.

Explanation

Correct: Filtering ensures that problematic text does not enter the RAG system, maintaining quality and ethical standards.

3. Rely on the LLM to ignore inappropriate language during processing.

Explanation

Incorrect: LLMs cannot inherently differentiate or exclude problematic content without explicit instructions.

4. Replace inappropriate words with placeholders in the ingested documents.

Explanation

Incorrect: Placeholder replacements may still retain some problematic context.

Overall explanation

Pre-ingestion filtering removes problematic text, ensuring the dataset's quality and ethical compliance.

Question 45

A Generative AI Engineer must identify source documents to train a RAG system for answering technical support questions. What factors should they prioritize?

1. **Relevance and quality of content based on the system's technical support requirements.**

Explanation

Correct: High-quality, relevant content ensures the system retrieves accurate and useful answers.

2. Availability of free or public domain documents.

Explanation

Incorrect: Public availability does not ensure alignment with technical support needs.

3. Diversity of sources over content relevance.

Explanation

Incorrect: Relevance to the task outweighs the need for diverse but unrelated sources.

4. Length of documents to maximize training data.

Explanation

Incorrect: Length alone does not guarantee relevance or quality.

Overall explanation

Selecting high-quality, relevant documents ensures the system retrieves accurate answers aligned with technical support requirements.